
Version 2.3 Final

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Natural Math

A Unified Framework for Acyclic Spatial Growth Processes

Version 2.3 — Final

Scope Statement

Natural Math models growth processes that share four properties:

1. Tree-like or forest-like structure. The parent relation forms a directed forest with no cycles. What grows is a branching hierarchy, not a general network.
2. Local decision-making. Each growing tip acts on information within a finite neighborhood. No global coordination, no central planner, no non-local state.
3. Finite resource budget. Growth, sensing, and retreat all cost energy. When energy is exhausted, activity stops. This is a theorem, not an assumption.
4. Spatial embedding. Positions matter. Growth happens in three-dimensional integer space with exact rational geometry.

This covers plant roots and shoots, fungal mycelium, vascular networks, neural arborization, and certain infrastructure systems. It does not cover cyclic networks, systems requiring global equilibrium, non-spatial growth, or independent reproduction that produces indefinite population persistence. Those require different frameworks or future extensions.

What Version 2.3 adds: Empirically validated regime taxonomy (emergent behavioral classes predictable from parameters), fully characterized reproduction extension (mechanism confirmed, structural limits mapped), population-level dynamics (heritable variation, selection, and the finding that reproduction extends but does not escape the thermodynamic bound), four worked examples spanning botany, urban systems, mycology, and neuroscience, and comprehensive validation against degenerate limits, adversarial environments, inverse prediction, conservation laws, scaling, and null models.

Reading Guide

If you are... Read... To understand...

New to the framework Part 1: Intuitive Overview What it does and why, no math

Implementing the system Part 2: Formal Specification Every definition, rule, and invariant

Curious about design choices Part 3: Design Rationale Why synchronous? Why SENSE costs energy?

Trying to apply it Part 4: Worked Examples Step-by-step in specific domains

Predicting macroscopic behavior Appendix A: Regime Taxonomy What kind of growth will my parameters produce?

Modeling populations Appendix B: Reproduction Extension Multi-forest dynamics, heritable variation, structural limits

Setting safe parameter ranges Appendix C: Parameter Boundaries Degenerate limits, hard invariants, meaningful ranges

Building an efficient implementation Appendix D: Computational Scaling Complexity, data structures, parallelization

Part 1: Intuitive Overview

What Natural Math Models

Natural Math describes how branching structures grow, adapt, and halt—using only local information and a finite energy budget. Think of roots navigating soil, fungi exploring a forest floor, growth cones wiring a brain, or neurons finding targets. No single tip sees the whole picture. Each makes simple choices based on what's nearby. The large-scale shape—the branching pattern, the zones of dense exploration, the points of retreat—emerges from thousands of local decisions.

The Core Idea

At each moment, a growing tip faces one of four choices:

- EXTEND: Keep growing forward, or split into two branches. Costs energy, explores new territory.
- SENSE: Wait, adjust direction, probe the boundary. Costs less energy but doesn't move.
- RESTRICT: Stop permanently. Energy too low, or neighbor too close. This branch is done.
- CONSERVE: Pull back and return unused energy to where you came from. The parent can reallocate it.

These four choices, made locally and repeatedly, generate the full range of behaviors: growth, branching, adaptation, retraction, and halt.

The Site Lifecycle

Every site in the system moves through states in a predictable order. Each state has both an entry condition and an exit condition—no site is ever stuck without a path forward (even if that path is death).

State	Decision	Entry Condition	Exit Condition
Growth	EXTEND	Energy available, space ahead, gradient present	Low energy, proximity blocked, pressure-triggered retreat
Exploration	SENSE	Flat gradient, near a boundary, or waiting	Gradient appears, energy runs low, or conditions trigger retreat
Retreat	CONSERVE	High pressure or external hibernate signal	N/A — site is removed; energy returns to parent
Death	RESTRICT	Energy below threshold, or neighbor too close	None — site is permanently frozen

This lifecycle is complete for an individual growth structure. It captures the full arc from seed to final state.

Emergent Regimes

The same microscopic rules produce qualitatively different macroscopic behaviors depending on parameters. Three primary regimes emerge:

- Sparse: High $P_{\text{bifurcate}}$ (difficult branching). Structures are linear, deep, and narrow. Tips hit obstacles or boundaries before energy runs out. Coverage is low but depth is obstacle-limited rather than energy-limited.
- Balanced: Moderate $P_{\text{bifurcate}}$. Structures branch regularly while maintaining depth. This regime produces the most efficient exploration—good coverage without sacrificing reach.
- Bushy: Low $P_{\text{bifurcate}}$ (easy branching). Structures are dense, shallow, and wide. Depth saturates at a characteristic maximum (~95 lattice units under default parameters) regardless of energy. Energy goes to lateral coverage, not forward growth.

These regimes are predictable from input parameters alone (91% accuracy using $P_{\text{bifurcate}}$ and initial energy) and are recoverable from observed morphologies (~90–95% accuracy in inverse prediction). The regime structure is documented fully in Appendix A.

Reproduction: What It Does and Doesn't Do

The framework supports an optional reproduction extension. A site that has thoroughly explored its local territory—measured by declining expansion efficiency (novel positions discovered per energy spent)—can create an independent offspring seed at a new location.

What reproduction does:

- Creates independent offspring forests with heritable variation in parameters
- Extends the active phase (2–15× longer than single forests)
- Increases structural diversity (multiple forests of different regimes interacting through shared trails)
- Supports selection: populations drift toward parameters that improve short-term fitness

What reproduction does not do:

- Produce indefinite population persistence
- The birth/death ratio approaches but never reaches 1.0, even under extreme energy input ($I=50/\text{tip/step}$), maximally permissive parameters, and evolutionary timescales

Why: The reproduction transaction has an irreducible energy cost. The parent sacrifices its productive position plus pays overhead. The offspring starts from a single seed and must re-establish before collecting energy efficiently. This gap cannot be closed by parameter tuning or selection—it is structural.

Reproduction is a life-history strategy: it extends, diversifies, and adapts, but does not escape the thermodynamic bound that guarantees all processes eventually freeze. The full specification and empirical characterization are in Appendix B.

How Tips Decide

A tip senses only what's nearby—other tips within range, obstacles it contacts directly, and a memory trail left by previous growth. It compares energy levels (moving toward richer areas), avoids places already visited, and responds to accumulated pressure (frustration at obstacles that eventually triggers branching or breakthrough).

The Three Strategies at an Obstacle

When a tip hits an obstacle:

- Dither (SENSE): Wait and probe. Cheap (2 energy per step) but doesn't move. At a permanent obstacle, the tip eventually runs out of energy and stops.
- Push through (repeated EXTEND): Hit the obstacle repeatedly. Expensive (10 energy per hit) but pressure builds up. At a threshold, the tip either bifurcates around the obstacle or erodes through it.
- Retreat (CONSERVE): Give up on this direction, return leftover energy to the parent. Cheap (1 energy plus recovery loss) and preserves resources for other branches.

No central planner chooses. The tip's local energy and pressure determine the outcome.

Energy is Everything

Every action costs energy. The system starts with a fixed total. When it's gone, growth stops. Even CONSERVE loses a fraction of energy (no perfect recycling), so the system cannot oscillate forever. Closed systems always reach an inactive state. This is a mathematical guarantee that has been empirically verified across all tested configurations. The same guarantee extends to populations of reproducing forests: birth rate never sustainably exceeds death rate, so populations also eventually freeze.

If you add an external energy source—sunlight, rainfall, investment, trophic factors—the system can persist longer. The theorem tells you exactly when halting is guaranteed.

Validated Across Six Dimensions

The framework has been systematically tested and has passed:

Test Result

Internal consistency Energy is strictly conserved across all transactions

Boundary behavior Degenerate parameters produce exactly the failures predicted by the invariants

Environmental robustness Adversarial geometries cannot break the regime classifier or core invariants

Scientific falsifiability Parameters are recoverable from morphology with ~90–95% accuracy

Computational scaling Mathematically finite at any scale; practically feasible with standard optimizations

Comparative value Produces qualitatively superior structures to null models (random walks)

What Emerges

From these simple rules, across domains, you get:

- Branching networks that efficiently explore space
- Retraction and reallocation when resources are scarce
- Pressure-driven breakthroughs at obstacles
- Trail-based memory that avoids redundant exploration
- Distinct behavioral regimes predictable and recoverable from parameters
- Heritable variation and selection when reproduction is enabled
- Emergent pathfinding in maze-like environments
- Global patterns from purely local decisions

The framework does not design the shape. It grows it.

Part 2: Formal Specification

1. The Lattice

1.1 Definition

The ambient space is the integer lattice:

$$\Lambda = \mathbb{Z}^3$$

This is locally finite: any ball of finite radius contains finitely many points.

1.2 Position Function

Each site p has a unique position:

$$\text{pos}: \mathcal{A}_t \rightarrow \Lambda$$

Positions are not required to be unique across sites, but conflict resolution rules (Section 8) prevent two sites from occupying the same position after an update.

2. State Definition

2.1 Global State

$$S_t = (\mathcal{A}_t, \mathcal{E}_t, \mathcal{P})$$

where:

- $\mathcal{A}_t$ is a finite set of site identifiers.
- $\mathcal{E}_t: \mathcal{A}_t \rightarrow \mathbb{Q}_{\geq 0}$ assigns energy to each active site.
- $\mathcal{P}$ is the fixed parameter tuple (all values in $\mathbb{Q}_{\geq 0}$ unless noted).

2.2 Site State

Each site $p \in \mathcal{A}_t$ carries:

$$x_{p,t} = (\text{pos}_p, \text{dir}_p, \text{type}_p, \text{pressure}_p, \text{parent}_p, \text{forest}_p)$$

Field	Type	Description
pos_p	$\mathbb{Z}^3$	Position in lattice
dir_p	$\mathbb{Z}^3 \setminus \{0\}$	Direction vector
type_p	$\{\text{seed}, \text{tip}, \text{branch}, \text{sensor}, \text{inert}\}$	Behavioral type
pressure_p	$\mathbb{Q}_{\geq 0}$	Accumulated constraint
parent_p	$\mathcal{A}_t \cup \{\text{nothing}\}$	Parent site identifier
forest_p	$\mathbb{N}$	Forest identifier (for multi-forest populations)

The forest field is used only when the reproduction extension is active (Appendix B). In single-forest configurations, all sites share the same forest identifier.

2.3 Total Energy

$$E_{\text{tot}}(t) = \sum_p \in \mathcal{A}_t E(p, t)$$

Energy lost to dissipation is not tracked separately; it is the difference $E_{\text{tot}}(0) - E_{\text{tot}}(t)$. Energy conservation has been empirically verified: total energy equals initial energy plus external input minus cumulative dissipation, with zero unaccounted loss or gain.

2.4 System Invariants

The following hold for all $t \geq 0$:

I1 (Parent acyclicity): The parent relation forms a directed forest with edges from child to parent. For all $p \in \mathcal{A}_t$, if $\text{parent}_p \neq \text{nothing}$, then parent_p was created at a strictly earlier timestep than p . Therefore there are no cycles.

I2 (Inert energy): For all $p \in \mathcal{A}_t$ with $\text{type}_p = \text{inert}$, $E(p, t) = 0$.

I3 (RESTRICT energy bound): For all $p \in \mathcal{A}_t$ that chose RESTRICT at $t-1$, either $E(p, t) < \tau$ (energy trigger) or $\min Q < \iota^2$ was true (proximity trigger). In both cases, RESTRICT sets $E(p, t) = 0$ and $\text{type}_p = \text{inert}$, satisfying I2.

I4 (Active sites only): Only sites with $\text{type}_p \in \{\text{seed}, \text{tip}, \text{branch}, \text{sensor}\}$ are evaluated by the decision function. Inert sites are never evaluated.

I5 (Finite support): $|\mathcal{A}_t|$ is finite for all t . The trail map has finite support at all t .

3. Neighborhood

3.1 Quadrance

For positions $u, v \in \mathbb{Z}^3$:

$$Q(u, v) = (u_x - v_x)^2 + (u_y - v_y)^2 + (u_z - v_z)^2 \in \mathbb{Z}_{\geq 0}$$

3.2 Neighborhood Definition

For site $p \in \mathcal{A}_t$ with integer quadrance bound $R \in \mathbb{Z}_{>0}$:

$$\mathcal{N}_R(p, t) = \{q \in \mathcal{A}_t \setminus \{p\} : Q(\text{pos}_p, \text{pos}_q) \leq R\}$$

The maximum possible neighborhood size is bounded:

$$|\mathcal{N}_R(p, t)| \leq N_{\max}(R) = |\{x \in \mathbb{Z}^3 : Q(x, \mathbf{0}) \leq R\}|$$

This is finite for all R , guaranteeing bounded computation per site per timestep. For $R = 25$, $N_{\max} \approx 523$; for $R = 30$, $N_{\max} \approx 739$.

3.3 Co-located Sites

If $Q(\text{pos}_p, \text{pos}_q) = 0$ for $p \neq q$, both sites are at the same lattice point. For gradient computation, co-located sites are included in the sum; the denominator $Q(p, q) = 0$ is replaced by 1 as a regularization convention.

4. Rational Geometry

4.1 Spread

For nonzero vectors $u, v \in \mathbb{Z}^3$:

$$s(u, v) = \frac{Q(u \times v, \mathbf{0})}{Q(u, \mathbf{0}) \cdot Q(v, \mathbf{0})}$$

where $u \times v$ is the integer cross product. This gives a rational number in $[0, 1]$. Geometrically, $s = \sin^2(\theta)$ where θ is the angle between u and v .

4.2 Bifurcation Spread Constraint

For parent direction v_0 and child directions v_1, v_2 :

$$s(v_0, v_1) = s, \quad s(v_0, v_2) = s, \quad s(v_1, v_2) = 4s(1-s)$$

where $s \in \mathbb{Q} \cap (0, 1)$ is the bifurcation spread parameter. The constraint $s(v_1, v_2) = 4s(1-s)$ follows from $\sin^2(2\theta) = 4\sin^2(\theta)\cos^2(\theta)$: the angle between children is twice the angle from parent to each child.

4.3 Constructing Child Directions

Given v_0 and s , pick a perpendicular vector w such that $s(v_0, w) = 1$ (i.e., $w \perp v_0$). Then set:

$$v_1 = v_0 + k \cdot w, \quad v_2 = v_0 - k \cdot w$$

where $k \in \mathbb{Q}$ is chosen so that $s(v_0, v_1) = s$. Solving:

$$k^2 = \frac{s}{1-s} \cdot \frac{Q(v_0, \mathbf{0})}{Q(w, \mathbf{0})}$$

If a rational k exists, use it. If not, scale v_0 and w by denominators until k is rational. This always terminates (multiply by denominators of $s/(1-s)$ and the quadrance ratio). The resulting v_1, v_2 are integer vectors satisfying the spread constraints exactly.

4.4 Direction Update (Bending)

For current direction $v_t \in \mathbb{Z}^3 \setminus \{0\}$, target direction $v_{\text{target}} \in \mathbb{Z}^3 \setminus \{0\}$, and bend strength $\alpha \in \mathbb{Z}_{\geq 0}$:

$$v_{t+1} = v_t + \alpha \cdot v_{\text{target}}$$

This is an integer vector. The default is $\alpha = 1$.

Position update on EXTEND:

$$\text{pos}_{\text{new}} = \text{pos}_p + v_{t+1}$$

5. Decision States

5.1 State Set

$$T_{\{p,t\}} \in \{+1, 0, -1, -2\}$$

Value Name Meaning

+1 EXTEND Grow or bifurcate

0 SENSE Adapt, redirect, remain

-1 RESTRICT Halt permanently, lose all energy

-2 CONSERVE Retract, return energy to parent

When the reproduction extension is active (Appendix B), a fifth state is added: $T = -3$ (REPRODUCE).

5.2 Gradient Estimation

For $p \in \mathcal{A}_t$:

$$\nabla \mathcal{E}(p,t) = \sum_{q \in \mathcal{N}_R(p,t)} \frac{E(q,t) - E(p,t)}{\max(Q(\text{pos}_p, \text{pos}_q), 1)} \cdot (\text{pos}_q - \text{pos}_p)$$

The trail gradient:

$$\nabla \text{Trail}(p,t) = \sum_{q \in \mathcal{N}_R(p,t)} \frac{\text{Trail}(\text{pos}_q) - \text{Trail}(\text{pos}_p)}{\max(Q(\text{pos}_p, \text{pos}_q), 1)} \cdot (\text{pos}_q - \text{pos}_p)$$

The effective gradient:

$$\nabla_{\text{eff}}(p,t) = \nabla \mathcal{E}(p,t) - \gamma \cdot \nabla \text{Trail}(p,t)$$

5.3 Gradient Magnitude

$$|\nabla_{\text{eff}}(p,t)|_Q = Q(|\nabla_{\text{eff}}(p,t)|, \mathbf{0})$$

5.4 Empty Neighborhood

If $\mathcal{N}_R(p,t) = \varnothing$:

$$\nabla_{\text{eff}}(p,t) = \mathbf{0}, \quad |\nabla_{\text{eff}}(p,t)|_Q = 0$$

5.5 Decision Function (Mutually Exclusive)

Conditions are evaluated in strict priority order.

Condition 1 — CONSERVE ($T = -2$)

$$(\text{parent}_p \neq \varnothing) \wedge (\text{parent}_p \in \mathcal{A}_t) \wedge \big(\text{pressure}(p,t) > P_{\text{hibernate}} \vee \text{external_hibernate_signal}(p,t) = \text{true} \big)$$

The hibernate pressure threshold defaults to $P_{\text{hibernate}} = 0.8 \cdot P_{\text{erode}} = 40$. The `external_hibernate_signal` is a domain-specific hook; defaults to `false`.

Condition 2 — RESTRICT ($T = -1$)

$$(E(p,t) < \tau) \vee \left(\min_{q \in \mathcal{N}_R(p,t)} Q(\text{pos}_p, \text{pos}_q) < \iota^2 \right)$$

But only if Condition 1 is false. The inequality $\min Q < \iota^2$ is strict. Exact equality does not trigger RESTRICT.

On RESTRICT: $E(p, t+1) = 0$, type_p becomes inert. The site permanently halts and all remaining energy is abandoned.

Condition 3 — EXTEND ($T = +1$)

$(E(p, t) \geq \tau) \wedge \left(\min_{q \in \mathcal{N}_R(p, t)} Q(\text{pos}_p, \text{pos}_q) > \iota^2 \right) \wedge (\|\nabla_{\text{eff}}(p, t)\|_Q > \eta^2)$

Additional constraint: If $\text{pressure}(p, t) \geq P_{\text{bifurcate}}$, require $E(p, t) \geq \epsilon_{\text{split}} + 2\tau$. This ensures both children have energy at least τ to survive one step.

If pressure triggers bifurcation but energy is insufficient, EXTEND is false; the site falls through to Condition 4.

Condition 4 — SENSE ($T = 0$)

All remaining cases.

Within SENSE, the site may adapt its behavior:

- If $\iota^2 < \min Q \leq (\iota + \delta)^2$: the site is in the approach zone. It may reduce its direction magnitude and/or adjust direction away from the nearest neighbor.
- If $\|\nabla_{\text{eff}}\|_Q \leq \eta^2$: the gradient is flat. The site may perturb its direction to explore the local energy landscape.
- Otherwise: the site maintains its current direction but does not extend.

SENSE-to-CONSERVE fallback: If $T = 0$ and $E(p, t) - \epsilon_{\text{sense}} < \tau$, and $\text{parent}_p \neq \text{nothing}$ and $\text{parent}_p \in \mathcal{A}_t$, then $T = -2$ (CONSERVE). This prevents sites from wasting their last energy on a SENSE that would trigger RESTRICT next step. The fallback applies in the narrow window $\tau \leq E < \tau + \epsilon_{\text{sense}}$.

5.6 Decision Function Parameters

Symbol Meaning Domain

τ Energy threshold $\mathbb{Q}_{\geq 0}$

ι Contact inhibition distance $\mathbb{Q}_{>0}$

η^2 Squared gradient threshold $\mathbb{Q}_{\geq 0}$

δ Sense boundary width $\mathbb{Q}_{\geq 0}$

γ Trail avoidance weight $\mathbb{Q}_{\geq 0}$

6. Energy Dynamics

6.1 Energy Cost Function

$\kappa(T) = \begin{cases} \epsilon_{\text{extend}}, & T = +1 \\ \epsilon_{\text{sense}}, & T = 0 \\ E(p, t), & T = -1 \\ \epsilon_{\text{conserve}}, & T = -2 \end{cases}$

All costs are in $\mathbb{Q}_{\geq 0}$. On RESTRICT ($T = -1$), the cost is the site's entire remaining energy; the site is set to $E = 0$.

Critical requirement for eventual inactivity: $\epsilon_{\text{extend}} > 0$ and $\epsilon_{\text{sense}} > 0$. Define:

$\epsilon_{\min} = \min(\epsilon_{\text{extend}}, \epsilon_{\text{sense}}) > 0$

These strict inequalities are hard invariants. Violating them (setting either cost to zero) breaks the termination proof and produces non-terminating behavior.

When the reproduction extension is active (Appendix B), an additional cost $\epsilon_{\text{reproduce}}$ applies to $T = -3$.

6.2 Standard Energy Update

For $T_{\{p,t\}} \in \{+1, 0, -1\}$:

$$E(p, t+1) = E(p, t) - \kappa(T_{\{p,t\}})$$

For $T = -1$ (RESTRICT), this becomes $E(p, t+1) = 0$.

6.3 CONSERVE Update

For $T_{\{p,t\}} = -2$, let $q = \text{parent}_p$:

Phase 2 of the update processes CONSERVE sites in topological order (tips toward root). A temporary energy buffer $E_{\{\text{buffer}\}}$ is used:

$$E_{\{\text{buffer}\}}(q) = E(q, t) + \beta \cdot (E(p, t) - \varepsilon_{\{\text{conserve}\}})$$

$$E(p, t+1) = 0$$

$$p \notin \mathcal{A}_{t+1}$$

$$\text{pressure}_{\{\text{buffer}\}}(q) = \text{pressure}(q, t) + \beta \cdot \text{pressure}(p, t)$$

where $\beta \in (0, 1)$. The strict inequality $\beta < 1$ is a hard invariant—it ensures net energy loss on every CONSERVE transaction.

Phase 5 applies costs using buffered values. If multiple children CONSERVE to the same parent, buffer accumulations are additive. A parent that itself CONSERVEs uses its buffered energy for its own transfer. Since topological ordering processes children before parents, a parent's CONSERVE always includes any child contributions.

Pressure transfers are scalar. The receiving parent applies transferred pressure to its own local context, not the child's. A child blocked by a stone at position X passes frustration magnitude to the parent, which applies it to whatever obstacle the parent faces at its own position.

6.4 Bifurcation Energy Split

Parent site p splits into children c_1, c_2 . Parent energy becomes 0:

$$E(c_1, t+1) = E(c_2, t+1) = \frac{E(p, t) - \varepsilon_{\{\text{split}\}}}{2}$$

Requires $E(p, t) \geq \varepsilon_{\{\text{split}\}}$. Children inherit $\text{parent} = p$.

If gradient bias is used:

$$E_{\{\text{closer}\}} = \frac{E(p,t) - \varepsilon_{\{\text{split}\}}}{2} + \Delta E_{\{\text{bias}\}}$$

$$E_{\{\text{farther}\}} = \frac{E(p,t) - \varepsilon_{\{\text{split}\}}}{2} - \Delta E_{\{\text{bias}\}}$$

6.4a Single Extension Energy

When a site EXTENDs without bifurcation (single child), parent energy becomes 0:

$$E_{\{\text{child}\}} = E_{\{\text{parent}\}} - \varepsilon_{\{\text{extend}\}} - \varepsilon_{\{\text{spawn}\}}$$

where $\varepsilon_{\{\text{spawn}\}} \in \mathbb{Q}_{\geq 0}$ is the overhead for creating any child. Default: $\varepsilon_{\{\text{spawn}\}} = \varepsilon_{\{\text{split}\}} / 2$.

6.5 Conflict Loser Energy

All energy costs are applied in Phase 5 based on each site's final decision after conflict resolution. A site that originally chose EXTEND in Phase 1 but was forced to SENSE by a conflict pays only the SENSE cost:

$$E(\text{loser}, t+1) = E(\text{loser}, t) - \epsilon_{\text{sense}}$$

6.6 Open-System Extension

$$E(p, t+1) = E(p, t) - \kappa(T_{p,t}) + I(p, t) + \sum_{q \in \mathcal{N}_R(p,t)} J_{q \rightarrow p, t}$$

$I(p,t)$ is external input. $J_{q \rightarrow p, t}$ is local transfer satisfying $J_{q \rightarrow p, t} = -J_{p \rightarrow q, t}$. The eventuality theorem applies only when $I(p,t) = 0$ for all p, t .

7. Pressure Dynamics

7.1 Pressure Accumulation

Pressure accumulates from two sources, both contributing to a single accumulator:

Obstacle contact: When an EXTEND candidate is rejected due to $\text{Env}(x) = \text{obstacle}$:

$$\text{pressure}(p, t+1) = \text{pressure}(p, t) + \Delta P_{\text{contact}}$$

Conflict loss: When an EXTEND candidate loses a spatial conflict to another site:

$$\text{pressure}(\text{loser}, t+1) = \text{pressure}(\text{loser}, t) + \Delta P_{\text{conflict}}$$

The site does not distinguish between obstacle pressure and competition pressure.

On RESTRICT, pressure resets to 0. The site's accumulated constraint history is discarded—it has given up entirely. On CONSERVE, pressure transfers to the parent (Section 6.3). This asymmetry is intentional: CONSERVE preserves experiential information; RESTRICT abandons it.

7.2 Pressure-Driven Bifurcation

If $\text{pressure}(p, t) \geq P_{\text{bifurcate}}$ and $T_{p,t} = +1$, bifurcation is triggered with spread:

$$s = s_{\min} + (s_{\max} - s_{\min}) \cdot \min\left(1, \frac{\text{pressure}(p,t)}{P_{\text{erode}}}\right)$$

7.3 Pressure-Driven Erosion

If $\text{pressure}(p, t) \geq P_{\text{erode}}$:

$$\text{Env}(x) \rightarrow \text{empty} \quad \text{where } x \text{ is the contact position}$$

$$\text{pressure}(p, t+1) = 0$$

Erosion occurs at the contact position. The parent's next EXTEND attempt at that position will succeed.

7.4 Pressure Parameters

Symbol Meaning Default

$\Delta P_{\text{contact}}$ Pressure per obstacle hit 5

$\Delta P_{\text{conflict}}$ Pressure per lost conflict 2

$P_{\{\text{bifurcate}\}}$ Threshold for branching 20
 $P_{\{\text{erode}\}}$ Threshold for erosion 50
 $P_{\{\text{hibernate}\}}$ Threshold for CONSERVE 40
 $s_{\{\min\}}$ Minimum spread 1/16
 $s_{\{\max\}}$ Maximum spread 1/2

8. Update Execution Sequence

The update proceeds in ordered phases.

Phase 1: Decision Computation

For all $p \in \mathcal{A}_t$, compute $T_{\{p,t\}}$ using the decision function (Section 5.5) evaluated on S_t . When the reproduction extension is active, the decision function includes Condition 2.5 (REPRODUCE) as specified in Appendix B.

Phase 2: CONSERVE Processing

Process sites with $T_{\{p,t\}} = -2$ in topological order from tips toward root:

- A site p may CONSERVE only if no child of p that also chose CONSERVE is still unprocessed.
- Transfer energy and pressure to parent_p via the buffer mechanism (Section 6.3).
- Mark p for removal from $\mathcal{A}_{t+1}$.

Phase 3: Candidate Generation

For all p with $T_{\{p,t\}} = +1$, compute candidate new positions:

- Extension: candidate position = $\text{pos}_p + \text{dir}_p$. New site type = tip; p becomes type branch.
- Bifurcation: two candidate positions = $\text{pos}_p + v_1$, $\text{pos}_p + v_2$, with v_1 , v_2 satisfying spread constraints (Section 4.3). Two new tip sites; p becomes type branch.

Parent energy becomes 0 in both cases.

Phase 4a: Obstacle Rejection

For each candidate position x :

- If $\text{Env}(x) = \text{obstacle}$: reject the candidate. Mark the parent as "obstacle-blocked."
- If $\text{Env}(x) = \text{target}$: candidate succeeds. Mark parent as "target-reached." Domain-specific hooks may override the parent's subsequent behavior (e.g., transition to sensor type for synapse formation).

Obstacle-blocked parents: Remain EXTEND for this timestep, pay $\epsilon_{\{\text{extend}\}}$, accumulate $\Delta P_{\{\text{contact}\}}$, do not move. Children are not spawned.

Phase 4b: Site-Site Conflict Resolution

For all remaining candidate positions within quadrance ι^2 of each other or within ι^2 of an existing site position:

- The candidate with lower parent energy yields.
- If equal energy, lower parent site identifier yields.
- Yielding candidate: parent switches to SENSE, pays $\epsilon_{\{\text{sense}\}}$, accumulates $\Delta P_{\{\text{conflict}\}}$.
- Winning candidate: proceeds to Phase 5.

Phase 5: State Application

- Apply energy costs for all sites according to their final decisions, using buffered values for sites that received CONSERVE transfers.
- Process REPRODUCE sites if the extension is active (Appendix B).
- Add successfully spawned children to $\mathcal{A}_{t+1}$.
- Update positions and directions for extended/bifurcated parents.
- Remove CONSERVE and REPRODUCE sites from $\mathcal{A}_{t+1}$.
- Sites with $T = -1$ remain in $\mathcal{A}_{t+1}$ with type inert and zero energy.

Phase 6: Trail and Pressure Updates

- Apply trail deposition at all active (non-inert) positions in $\mathcal{A}_{t+1}$.
- Apply trail decay globally.
- Apply pressure updates.

9. Type System

9.1 Type Set

$\text{Type} = \{\text{seed}, \text{tip}, \text{branch}, \text{sensor}, \text{inert}\}$

9.2 Type Transition Function

Current type EXTEND (+1) SENSE (0) RESTRICT (-1) CONSERVE (-2)

seed branch sensor inert removed

tip branch sensor inert removed

branch branch sensor inert removed

sensor tip sensor inert removed

inert inert inert inert removed

Note: After EXTEND, parent type becomes branch with $E = 0$. Branch sites do not spawn additional children in the base framework. Branch reactivation (lateral tip spawning) is an open extension (Open Question #3).

When the reproduction extension is active, REPRODUCE (-3) follows the same transition as CONSERVE: site is removed from $\mathcal{A}_{t+1}$.

9.3 Spawn Rules

Parent action Children spawned Child types

EXTEND (tip, sensor) 1 child tip

EXTEND (seed) 1 child tip

EXTEND + bifurcation 2 children tip, tip

EXTEND (branch) 0 children —

9.4 Site Fate on Termination

Termination Mode Type Change In $\mathcal{A}_{t+1}$? Energy

RESTRICT becomes inert yes zero

CONSERVE deleted no N/A

10. Environment

10.1 Environment Map

$\text{Env}: \mathbb{Z}^3 \rightarrow \{\text{empty}, \text{obstacle}, \text{target}\}$

Initialized before simulation. Modified only by pressure-driven erosion (Section 7.3).

10.2 Obstacle Contact

Agents do not have direct access to Env. They sense obstacles only through contact during an EXTEND attempt, as specified in Phase 4a (Section 8).

11. Environmental Trails

11.1 Trail Map

$\text{Trail}: \mathbb{Z}^3 \rightarrow \mathbb{Q}_{\geq 0}$

Initialized to zero. Stored as a sparse map in implementation. Support is finite at all t.

11.2 Deposition

For each $p \in \mathcal{A}_{t+1}$ with $\text{type}_p \neq \text{inert}$:

$$\text{Trail}(\text{pos}_p) \leftarrow \text{Trail}(\text{pos}_p) + \Delta_{\text{trail}}$$

Inert sites do not deposit trails. They are structural remnants, not active agents.

When multiple forests share the same lattice (reproduction extension), all forests deposit to and read from the same trail map. Trail-based interaction between forests is the primary mechanism of inter-forest competition.

11.3 Decay

$$\text{Trail}(x) \leftarrow \max(0, \text{Trail}(x) - \Delta_{\text{decay}}) \quad \text{forall } x \in \mathbb{Z}^3$$

In practice, decay is applied only to the finite support set of Trail.

11.4 Parameters

Symbol	Meaning	Default
Δ_{trail}	Trail per visit	0.5
Δ_{decay}	Decay per timestep	0.01
γ	Avoidance weight	1.0

12. Master Update (Two-Phase Form)

Phase A — Decision:

$$T_{p,t} = \Theta(x_{p,t}, E(p,t), \mathcal{N}_R(p,t), \mathcal{P}) \quad \text{forall } p \in \mathcal{A}_t$$

Phase B — Execution:

$$S_{t+1} = \Psi(S_t, \{T_{p,t}\}_{p \in \mathcal{A}_t})$$

where Ψ encompasses Phases 2–6 from Section 8.

13. Eventual Inactivity Theorem

Statement

Let a Natural Math process satisfy:

- $E_{\text{tot}}(0) < \infty$.
- $I(p,t) = 0$ for all p, t (no external energy).
- $\epsilon_{\text{extend}} > 0$ and $\epsilon_{\text{sense}} > 0$.
- $\beta \in (0,1)$ (CONSERVE loses energy).
- Energy transfers $J_{q \rightarrow p}$ are conservative.

Then there exists $T < \infty$ such that for all $t \geq T$ and all $p \in \mathcal{A}_t$:

$$T_{p,t} \in \{-1\} \text{ or } p \text{ is inert}$$

The system reaches a frozen state: no further EXTEND, SENSE, or CONSERVE decisions occur. RESTRICT and inert sites may remain, but the active process has halted.

Proof

Define $E_{\text{tot}}(t) = \sum_{p \in \mathcal{A}_t} E(p, t)$.

EXTEND steps: Each consumes $\epsilon_{\text{extend}} \geq \epsilon_{\min} > 0$.

SENSE steps: Each consumes $\epsilon_{\text{sense}} \geq \epsilon_{\min} > 0$.

CONSERVE steps: Net change $\Delta E = -(1-\beta)E_{\text{child}} - \beta\kappa_{\text{conserve}} < 0$ since $\beta < 1$ and $E_{\text{child}} \geq 0$.

RESTRICT steps: $E(p, t+1) = 0$. Energy decreases by $E(p,t) \geq 0$. These steps are not counted in the active timestep bound.

Transfers J : Sum to zero, no effect on E_{tot} .

Conflicts: Losers pay $\epsilon_{\text{sense}} \geq \epsilon_{\min}$.

In any timestep where at least one EXTEND, SENSE, or CONSERVE occurs, or any conflict produces a SENSE, total energy decreases by at least $\epsilon_{\min}$. RESTRICT-only timesteps do not count; they represent the system entering its frozen state. Invariants I2 and I3 ensure inert and RESTRICT sites hold no energy that could fuel further activity.

Let N_{active} be the number of timesteps with at least one such energy-decreasing event. Since $E_{\text{tot}}(t) \geq 0$ and $E_{\text{tot}}(0)$ is finite:

$$N_{\text{active}} \leq \left\lfloor \frac{E_{\text{tot}}(0)}{\epsilon_{\min}} \right\rfloor$$

After at most this many active timesteps, no EXTEND, SENSE, or CONSERVE can occur. All remaining sites must have $T = -1$ or be inert. The system is frozen. $\square$

Empirical verification: Energy conservation has been confirmed in simulation across closed, open, and adversarial configurations. Total energy equals initial energy plus external input minus cumulative dissipation with zero unaccounted loss or gain.

Population-level extension: When the reproduction extension is active, each REPRODUCE event pays a fixed cost $\epsilon_{\text{reproduce}}$ and creates a new seed with energy E_{seed} drawn from the parent. The net energy change is $-(E_{\text{seed}} + \epsilon_{\text{reproduce}})$ for the parent forest plus $+E_{\text{seed}}$ for the offspring forest, with a net system-wide loss of at least $\epsilon_{\text{reproduce}} > 0$. The total energy across all forests therefore continues to decrease monotonically. Empirically, populations of reproducing forests always freeze, even under high external energy input ($I \leq 50$), maximally permissive parameters, and evolutionary timescales. The birth/death ratio approaches but does not reach 1.0. See Appendix B for full characterization.

14. Field Formulation

$$\mathfrak{F}: \mathbb{Z}^3 \times \mathbb{N} \rightarrow \{\text{active}, \text{inert}, \text{empty}\} \times \mathbb{Q}_{\geq 0} \times \text{Type} \times \mathbb{Z}^3$$

The update:

$$\mathfrak{F}(\cdot, t+1) = \begin{cases} \mathfrak{F}(\cdot, t), & \text{if all sites at } t \text{ are inert or empty} \\ \Psi(\mathfrak{F}(\cdot, t)), & \text{otherwise} \end{cases}$$

15. Open Questions

The following must be specified per application:

1. External hibernate signal: What domain-specific conditions trigger CONSERVE beyond pressure? (Seasonal cues, policy changes, resource shocks.)
2. Bifurcation trigger beyond pressure: Does energy surplus alone trigger branching? Default: no.
3. Branch spawning: When does a branch site spawn a lateral tip? Default: never. Note that branches have $E = 0$ after spawn; any branch reactivation would require external energy input or a redistribution mechanism.
4. Trail decay rate: Domain-dependent.
5. Obstacle representation: How are obstacles stored and queried?
6. Gradient bias ΔE_{bias} : Default is 0.
7. Exploration drive: Optional n_{patience} parameter forces EXTEND after n consecutive SENSE steps. Default: ∞ (disabled).
8. Population persistence mechanisms: Reproduction (Appendix B) extends the active phase and enables heritable variation, but does not produce indefinite population persistence under any tested configuration. Mechanisms that might enable persistence—lateral spread (runners/stolons), multi-tip seeds, inter-forest energy transfer—are beyond the current framework scope and reserved for future investigation.
9. Target-triggered state transitions: When a tip contacts a target site ($\text{Env}(x) = \text{target}$), what state transition occurs? Neural: synapse formation (transition to sensor type). Root: growth cessation (inert). Fungal: penetration or surface attachment. Domain-specific hook using the existing Phase 4a Env check.

Part 3: Design Rationale

Why Synchronous Evaluation?

All sites evaluate their decisions based on the same pre-update state S_t . Alternatives considered:

- Asynchronous (random order): Outcome depends on update order. Two runs with identical initial state can diverge. Termination proof becomes order-dependent.
- Continuous time: Requires differential equations; loses discrete computational guarantee.

Synchronous evaluation guarantees consistent information. The two-phase design eliminates circular dependencies. All Phase 1 decisions are independent and parallelizable.

Why Topological Ordering for CONSERVE?

When multiple sites retract, a child might CONSERVE to its parent, and that parent might also CONSERVE. Processing children before parents ensures the parent receives the child's energy before deciding whether to retract. This matches biological intuition—retraction waves move from tips toward the base.

Why Does SENSE Cost Energy?

Three reasons:

1. Physical: Maintenance costs energy even when not growing. A root tip is metabolically active. A growth cone is probing. A company's R&D department burns budget while exploring options.
2. Mathematical: If SENSE were free, the termination proof would fail—a system could SENSE forever. Making SENSE cost energy is the simplest way to guarantee finite behavior without additional assumptions.
3. Design: Positive SENSE cost creates a meaningful tradeoff. Extended sensing has opportunity cost—energy spent sensing is not spent extending. This forces balance between exploration and exploitation.

Why the 5:1 Cost Ratio?

The defaults ($\epsilon_{\text{extend}} = 10$, $\epsilon_{\text{sense}} = 2$) encode three strategies at an obstacle:

- Dither (SENSE): Cheap (2/step) but ineffective—never moves. Slow death at permanent obstacles.
- Push through (repeated EXTEND): Expensive (10/attempt, 40 energy to reach $P_{\text{bifurcate}} = 20$, plus bifurcation overhead ~55–60 total). Effective—triggers branching or erosion.
- Retreat (CONSERVE): Cheap (1 plus recovery loss). Abandons the direction; reallocates energy to alternatives.

No central planner chooses. Local energy and pressure determine the outcome.

Why RESTRICT Abandons All Energy?

On RESTRICT, a site's energy is set to zero—it is not returned to the parent. This is in contrast to CONSERVE, which returns a β fraction. The asymmetry is intentional:

- RESTRICT: "I am dead. Nothing to pass on." The site hit a hard limit (energy depletion or crowding) and perished in place. Its remaining energy is lost to the environment.
- CONSERVE: "I am retreating strategically. Here is what I have left." The site recognized unfavorable conditions and pulled back, preserving resources for the rest of the structure.

This preserves Invariant I2 (inert sites have zero energy) and creates a meaningful strategic choice: retreat early to preserve resources, or push until the end and risk losing everything.

Why Scalar Pressure Cascades?

Pressure transfers from child to parent on CONSERVE carry magnitude only—not directional information about the source of blockage. A child blocked by a stone at position X passes frustration magnitude to the parent, which applies it to whatever obstacle the parent faces at its own position.

This is a deliberate simplification. It models "this branch is struggling" as a signal, without requiring the parent to know why or where. In biological terms: a root tip encountering a stone sends a biochemical signal upward; the parent root responds by trying harder in its own direction, not by targeting the child's stone.

Why the SENSE-to-CONSERVE Fallback?

A site with energy in the narrow window $\tau \leq E < \tau + \epsilon_{\text{sense}}$ that chooses SENSE would pay its SENSE cost, drop below τ , and RESTRICT on the next timestep—losing all remaining energy. The fallback intercepts this: if SENSE would push energy below τ , the site CONSERVES instead, returning a β fraction to the parent.

The window is small (default: $8 \leq E < 10$), but the energy preserved is meaningful relative to the total budget. The fallback encodes a simple heuristic: "If one more step of waiting would kill me, retreat now."

Why Parent Acyclicity is an Invariant

CONSERVE requires a unique parent. Topological ordering requires a partial order. Cycles would allow energy to circulate forever, violating the eventuality theorem. The invariant holds because children are only created by EXTEND from existing parents, and parents are never created by children.

Why Integer Lattice?

$\mathbb{Z}^3$ is locally finite by construction. This gives bounded neighborhood size ($N_{\max}$), exact integer position updates, and guaranteed finite computation per site per timestep. The cost is losing continuous direction resolution, but spread-based angle measurement on integer vectors preserves exact rational geometry.

Why Reproduction Is Structurally Dissipative

The reproduction transaction (Appendix B) has an irreducible cost structure:

- 1. Parent pays $E_{\text{seed}} + \epsilon_{\text{reproduce}}$ and becomes inert, losing all future productivity.
- 2. Offspring starts from a single seed and must EXTEND several times before collecting external energy efficiently.
- 3. The establishment phase is energy-negative even with external input, because a single tip collects I once per step while burning energy on EXTEND and SENSE.

This gap cannot be closed by parameter tuning, high external input, or selection. Empirically, the birth/death ratio approaches ~ 0.97 asymptotically as $I \rightarrow \infty$ under maximally permissive conditions, but never reaches 1.0. The population-level freeze is the natural extension of the individual-level eventuality theorem.

Part 4: Worked Examples

Example 1: Root Growth Through Soil

Domain Mapping

Framework Concept Root Domain
Agent Root tip (apical meristem)
Energy Stored carbohydrate
Obstacle Stone, compacted soil
Trail Depleted nutrient zone
Pressure Mechanical resistance
CONSERVE Seasonal dieback, stress abortion

Walkthrough

Timestep 1: Seed at (0,0,0), $E = 400$, direction (0,1,0), type = seed. Neighborhood empty. $|\nabla_{\text{eff}}|_Q = 0 \leq \eta^2$. Decision: SENSE. Cost: 2. Energy: 398.

Timestep 2: Same. Decision: SENSE. Cost: 2. Energy: 396.

Timestep 3: Target tropism provides gradient. Decision: EXTEND. Seed spawns child tip at (0,1,0). Seed becomes branch. Child energy: $396 - 10 - 7 = 379$.

Timestep 4: Tip at (0,1,0), $E = 379$. EXTEND to (0,2,0). New child energy: $379 - 17 = 362$.

Timestep 5: Tip at (0,2,0), $E = 362$. Stone at (0,3,0). Candidate rejected (Phase 4a). Parent remains EXTEND. Pays 10. Energy: 352. Pressure: 5.

Timestep 6: Tip at (0,2,0), $E = 352$, pressure = 5. $\min Q$ to parent = $1 = \eta^2$ (boundary case). Conditions 2 and 3 both fail. SENSE. Cost: 2. Energy: 350. Direction bent slightly.

Timestep 7: Tip at (0,2,0), $E = 350$, pressure = 5. EXTEND attempt offset from vertical. Still hits stone. Rejected. Pays 10. Energy: 340. Pressure: 10.

Timesteps 8–9: Repeated contact. Pressure reaches 15, then 20.

Timestep 10: Pressure = 20. $P \geq P_{\text{bifurcate}}$. EXTEND, bifurcation triggered. Two children with spread $s = 1/4$ ($\sim 60^\circ$), biased toward target. Children energy: $(340 - 14)/2 = 163$ each. Parent becomes branch.

Total bypass cost: 4 EXTEND attempts (40) + bifurcation overhead (14) + 2 SENSE steps (4) = 58 energy.

Example 2: Urban Development (Sketch)

Framework Concept Urban Domain
Agent Development zone
Energy Available capital
Obstacle Zoning restriction, protected land
Trail Market saturation
Pressure Land value pressure, density
CONSERVE De-zoning, land banking

Growth extends along transportation corridors. Zoning boundaries create obstacles. Pressure builds at restricted edges, triggering either policy change (erosion) or leapfrog development (bifurcation). Trail represents market saturation—recently developed areas have absorbed demand, steering growth toward underserved territory. Trail decay models demographic turnover, allowing eventual redevelopment.

Example 3: Fungal Mycelium (Sketch)

Framework Concept Fungal Domain
Agent Hyphal tip
Energy Internal nutrient reserves
Obstacle Air gap, toxic substrate
Trail Depleted substrate, self-recognition
Pressure Mechanical resistance
CONSERVE Hyphal retraction, resource reallocation

Hyphae EXTEND into nutrient-rich areas, SENSE at boundaries, RESTRICT when exhausted, and CONSERVE to retract exploratory hyphae. Trail avoidance (self-recognition) prevents self-crossing. Network fractal dimension emerges from branching spread and nutrient distribution.

Example 4: Neural Arborization

Domain Mapping

Framework Concept Neural Domain
Agent Growth cone (axon or dendrite tip)
Energy ATP, mitochondrial transport, trophic factors
Obstacle Repulsive cues (semaphorins, slits, myelin inhibitors)
Trail Self-avoidance signals (ephrin/Eph), depleted territory
Pressure Mechanical tension + inhibitory signal accumulation
CONSERVE Axon retraction, developmental pruning

Biological Fidelity

The neural domain maps onto the framework with exceptional fidelity (~85–90%). Growth cones are the biological realization of the tip/sensor distinction: highly motile, actin-rich structures that integrate local cues and make extension, turning, branching, or retraction decisions without global knowledge. The four decision states map directly: EXTEND is actin-driven protrusion, SENSE is filopodial exploration, RESTRICT is stalled or eliminated growth, and CONSERVE is active pruning with resource reallocation.

Regime Correspondence

The three primary regimes match distinct neural cell classes:

- Sparse (high $P_{\text{bifurcate}}$): Long-range projection axons (corticospinal tract, callosal projections). Minimal branching, depth-prioritized, obstacle-limited by guidance corridors.
- Balanced (moderate $P_{\text{bifurcate}}$): Cortical pyramidal cells, hippocampal CA1 neurons. Regular branching with efficient coverage and good depth.
- Bushy (low $P_{\text{bifurcate}}$): Purkinje cell dendrites, stellate interneurons. Dense local arborization with saturated depth and extensive lateral coverage.

Parameter Suggestions for Cortical Pyramidal Neuron

Parameter	Value	Rationale
τ	12	Higher ATP threshold for growth cone advance
ϵ_{extend}	10	High cytoskeletal cost (actin treadmilling, microtubule invasion)
ϵ_{sense}	2	Probing with filopodia is cheaper than full extension
β	0.8	Pruning recycles most but not all material
R	30	Filopodia reach ~10–50 μm
ι	2	Minimum spacing to avoid fasciculation overlap
α	2	Moderate bend strength toward chemoattractant gradients
η^2	0.5	Sensitive chemotaxis (netrin, BDNF, NGF detection)
γ	1.5	Strong self-avoidance for dendritic tiling
$P_{\text{bifurcate}}$	14–20	Balanced regime for pyramidal cells

Domain-Specific Hooks

Target recognition: When a growth cone contacts a target site ($\text{Env}(x) = \text{target}$ in Phase 4a), it transitions to sensor type—stabilized, metabolically active, but no longer extending. This models synapse formation without new decision states. The sensor type already exists and represents "alive but not advancing."

Activity-dependent external input: External input $I(p,t)$ (Section 6.6) can be modulated by recent EXTEND success and target proximity—both locally computable. This models trophic factor rescue (successful growth cones receive more support) and Hebbian refinement without violating local agency.

Walkthrough: Growth Cone at a Choice Point

A growth cone approaching a target field encounters a repulsive barrier (glial boundary, semaphorin gradient). It SENSEs at the barrier, accumulating pressure with each failed EXTEND attempt. At $P_{\text{bifurcate}}$, it bifurcates—one branch continuing toward the target along a permissive corridor, the other CONSERVEing and returning energy to the parent. The successful branch reaches the target, transitions to sensor type, and stabilizes. The abandoned branch's energy funds further exploration elsewhere. Total cost is comparable to the root example (~55–65 energy), but the outcome is a precisely targeted connection rather than a simple bypass.

Null Model Comparison

Random walk models (unlimited or with stochastic death) fail to reproduce neural arborization patterns. They produce tangled, self-overlapping structures with no obstacle navigation, no resource reallocation, and no regime structure. Natural Math's combination of gradient sensing, pressure-driven branching, trail-based self-avoidance, and CONSERVE reallocation is necessary for realistic neural morphologies.

Parameter Groups

Energy Budget

Parameter	Meaning	Default	Safe Range
τ	Minimum energy to extend	8	3–25
ϵ_{extend}	EXTEND cost	10	3–30
ϵ_{sense}	SENSE cost	2	1–8
$\epsilon_{\text{restrict}}$	RESTRICT cost	all remaining	—

$\backslash\text{varepsilon}_{\text{conserve}}$ CONSERVE cost 1 0.5–5
 $\backslash\text{varepsilon}_{\text{split}}$ Bifurcation overhead 14 5–30
 $\backslash\text{varepsilon}_{\text{spawn}}$ Overhead per child 7 3–15
 $\backslash\text{beta}$ CONSERVE recovery fraction 0.8 0.6–0.95

Geometry

Parameter Meaning Default Safe Range
R Neighborhood quadrance bound 25 10–50
 $\backslash\text{iota}$ Minimum spacing 1 1–5
 $\backslash\text{alpha}$ Bend strength 1 1–5
s Bifurcation spread 1/4 1/16–1/2

Pressure

Parameter Meaning Default Safe Range
 $\Delta P_{\text{contact}}$ Pressure per obstacle hit 5 1–15
 $\Delta P_{\text{conflict}}$ Pressure per lost conflict 2 1–10
 $P_{\text{bifurcate}}$ Threshold for branching 20 5–100
 P_{erode} Threshold for erosion 50 20–200
 $P_{\text{hibernate}}$ Threshold for CONSERVE 40 15–150
 $s_{\min}$, $s_{\max}$ Spread range under pressure 1/16, 1/2 —

Memory

Parameter Meaning Default Safe Range
 Δ_{trail} Trail per visit 0.5 0.1–2.0
 Δ_{decay} Decay per timestep 0.01 0.001–0.1
 γ Avoidance weight 1.0 0–4

Gradient

Parameter Meaning Default Safe Range
 η^2 Squared gradient threshold 1 0.1–10
 δ Sense boundary width 2 0–10

Optional Domain Parameters

Parameter Meaning Default
 ΔE_{bias} Gradient-aligned child bonus 0
 n_{patience} Max SENSE before forced EXTEND ∞

Appendix A: Regime Taxonomy

A.1 Overview

The same microscopic rules produce qualitatively different macroscopic behaviors depending on parameter settings. These "regimes" are emergent—they are not specified in the rules but arise from the interaction of energy costs, branching thresholds, and spatial constraints.

Three primary regimes have been identified and validated through systematic simulation (Batches 1–2, 26 runs across parameter space). A fourth (Linear) appears under high $P_{\text{bifurcate}}$ with specific obstacle configurations.

A.2 Regime Descriptions

Sparse Regime

- Parameter signature: High $P_{\text{bifurcate}}$ (>20), any E_0
- Branch density: Low (~ 0.3 – 0.4)
- Depth: Obstacle-limited rather than energy-limited. Removing obstacles dramatically increases depth. Depth grows slowly with E_0 .
- Coverage: Low. Tips explore narrow paths.
- Mechanism: Tips hit obstacles or boundaries before exhausting energy. Bifurcation is rare. CONSERVE is rare. Structures are linear and deep.

Balanced Regime

- Parameter signature: Moderate $P_{\text{bifurcate}}$ (14–20), moderate-to-high E_0
- Branch density: Moderate (~ 0.4 – 0.5)
- Depth: Scales with E_0 and obstacle configuration. Not saturated.
- Coverage: Efficient. Good balance between exploration breadth and depth.
- Mechanism: Branching occurs regularly but doesn't consume the entire energy budget. Depth and coverage both benefit from increased energy.

Bushy Regime

- Parameter signature: Low $P_{\text{bifurcate}}$ (<14), sufficient E_0
- Branch density: High (~ 0.5 – 0.6)
- Depth: Saturates at ~ 90 – 110 regardless of E_0 (confirmed at $E_0 = 800, 1600, 3200$). Energy goes to lateral coverage, not forward growth.
- Coverage: High. Extensive lateral exploration.
- Mechanism: Easy branching causes frequent bifurcation. Energy is consumed by lateral spread. Depth hits a characteristic maximum set by the interaction of bifurcation cost, spread angle, and contact inhibition.

Linear Regime

- Parameter signature: Very high $P_{\text{bifurcate}}$ (>25), obstacles present
- Branch density: Very low (~ 0.25)
- Depth: Obstacle-dominated. Strongly dependent on obstacle density.
- Coverage: Low. Single-path exploration.
- Mechanism: Branching is nearly absent. Growth follows a single path whose length is determined by obstacle spacing and energy.

A.3 Regime Classifier

A decision tree trained on input parameters alone predicts regime with 91% accuracy (10/11 correct on validation data):

```
...
IF initial_energy ≤ 325: → Sparse
ELSE IF P_bifurcate ≤ 13.5:
IF initial_energy ≤ 1300: → Balanced-Bushy boundary
ELSE: → Bushy
ELSE IF P_bifurcate ≤ 25: → Balanced
ELSE: → Linear
...
```

The primary determinant is $P_{\text{bifurcate}}$. Initial energy modulates the boundary between Balanced and Bushy at low $P_{\text{bifurcate}}$. The transition near $P_{\text{bifurcate}} = 19$ – 21 is sharp and energy-independent—a genuine phase boundary in parameter space.

A.4 Inverse Prediction

The regime taxonomy is bidirectional. Given an observed morphology (depth, branch density, coverage, runtime), the key parameters can be recovered:

- $P_{\text{bifurcate}} \approx 35 - (\text{branch_density} \times 50)$
- For Balanced regime: $E_0 \approx \text{max_depth} \times 5.5 + \text{coverage_factor}$

Recovery accuracy is ~90–95% for primary parameters. Boundary cases (transitional morphologies) are correctly flagged with appropriate confidence. The forward–backward consistency confirms the taxonomy reflects real phase-like structure in parameter space rather than arbitrary bins.

A.5 Per-Regime Scaling Laws

Where validated, per-regime predictors substantially outperform global equations:

Metric Global Error Regime-Local Error Improvement

Coverage efficiency 41.8% 8.7% (Sparse), 9.3% (Bushy) 4.5×

Runtime 27.3% 11.1% (Linear), 15.2% (Bushy) 2×

Branch density 12.3% 12.3% (same) None—global equation adequate

Depth does not have a single scaling law. It is determined by different physics in each regime: obstacle-limited (Sparse), saturated (Bushy), obstacle-path-length (Linear), and potentially energy-branching tradeoff (Balanced).

A.6 Boundary Blending

Near regime boundaries ($P_{\text{bifurcate}} \approx 19\text{--}21$), behavior is probabilistic. A sigmoid blending model:

$$P(\text{Sparse}) = \text{sigmoid}((P_{\text{bifurcate}} - 20) / \text{width})$$

produces predictions that weight the two adjacent regime models. The transition width is approximately 2 units of $P_{\text{bifurcate}}$. Near the Balanced/Bushy boundary, energy modulates the transition.

A.7 Null Model Distinction

Regime-specific morphologies are not reproducible by unbiased random growth. Null models (random walks with and without stochastic death) produce tangled, self-overlapping structures with no obstacle navigation, no resource reallocation, and no regime structure. Trail avoidance, pressure-driven decisions, and energy budgeting are necessary for realistic space-filling and obstacle navigation. The framework's complexity is justified.

A.8 Empirical Status

Finding Confidence Basis

Three primary regimes exist High 26 runs, consistent clustering

Classifier accuracy ~91% High 10/11 validation

Inverse prediction ~90–95% High Blind recovery test

Bushy depth saturation at $D_{\text{max}} \approx 95$ High Confirmed at $E_0 = 800, 1600, 3200$

Sparse depth is obstacle-limited High Zero-obstacle run shows dramatic depth increase

Linear depth is obstacle-density-dependent High Varying obstacle density produces proportional depth changes

Per-regime coverage predictors (9% error) High Validated on multiple runs per regime

Adversarial environment robustness High Checkerboard, funnel, maze, all-obstacle, periodic

Null model superiority High Random walk comparison

Per-regime depth models Medium Mechanism identified, equations partially fitted

Appendix B: Reproduction Extension

B.1 Status

Mechanism: Validated (Batch 3A). Reproduction creates viable independent offspring forests.

Heritable variation: Validated (Multi-seed experiment). Mutation on $P_{\text{bifurcate}}$ produces heritable diversity in branching strategy. Selection acts on this variation—populations drift toward parameters that improve short-term fitness.

Population persistence: Not observed under any tested configuration (Batches 4A–D, Evolutionary test). Birth/death ratio approaches ~ 0.97 asymptotically but never reaches or exceeds 1.0. Reproduction is a life-history strategy: it extends the active phase, increases diversity, and enables adaptation, but does not produce indefinite population persistence.

B.2 Decision State

A fifth decision value is added to the state set (Section 5.1):

$T = -3 \quad \text{REPRODUCE}$

Priority order (inserted between RESTRICT and EXTEND):

1. CONSERVE (-2)
2. RESTRICT (-1)
3. REPRODUCE (-3) $\leftarrow$ new
4. EXTEND (+1)
5. SENSE (0)

This placement means a site only reproduces if it is not hibernating and not dead. Reproduction is an alternative to continued local growth, not a response to failure.

B.3 Entry Condition

REPRODUCE is triggered when all of the following hold:

$(\text{parent}_p \neq \text{nothing}) \wedge (E(p, t) \geq E_{\text{reproduce}}) \wedge (\text{ExpansionEfficiency}(p, t) < \eta_{\text{reproduce}})$

where expansion efficiency is defined as:

$\text{ExpansionEfficiency}(p, t) = \frac{\text{NovelPositions}(p, t-w, t)}{\text{EnergySpent}(p, t-w, t)}$

NovelPositions is the count of distinct lattice positions first visited by p or its recent descendants within the window $[t-w, t]$. EnergySpent is the total energy consumed by p and its recent descendants in that window. Both are locally computable from the site's subtree state.

Parameters:

Symbol Meaning Default

$E_{\text{reproduce}}$ Minimum energy to reproduce 50

$\text{varepsilon}_{\text{reproduce}}$ Reproduction overhead cost 10

$\eta_{\text{reproduce}}$ Expansion efficiency threshold 0.1

w Efficiency window (timesteps) 20

$E_{\text{max_seed}}$ Maximum seed energy 100

σ Mutation standard deviation on $P_{\text{bifurcate}}$ 1.0

B.4 Action

When a site chooses REPRODUCE:

1. Compute dispersal position: $\text{pos}_{\text{seed}} = \text{pos}_p + \mathbf{d}$ where $\mathbf{d}$ is a dispersal vector. Default: random direction, distance drawn from a domain-specific distribution (e.g., uniform in $[10, 30]$ lattice units).

2. Compute seed energy: $E_{\text{seed}} = \min(E(p, t) - \epsilon_{\text{reproduce}}, E_{\text{max_seed}})$.
3. Create a new seed site at pos_{seed} with:
 - Energy: E_{seed}
 - Direction: inherited from parent or randomized
 - Parameters: parent parameters + $\text{calN}(0, \sigma)$ mutation on mutable parameters (minimum: $P_{\text{bifurcate}}$)
 - Forest identifier: new unique value
 - Parent: nothing (the new seed is the root of an independent forest)
4. Parent pays: $E(p, t+1) = E(p, t) - \epsilon_{\text{seed}} - \epsilon_{\text{reproduce}}$. Parent becomes inert.
5. The new seed is added to calA_{t+1} in Phase 5.

B.5 Multi-Forest Dynamics

Multiple forests share the same lattice, environment map, and trail map. They interact only through these shared resources:

- Trail competition: Trail deposited by forest A repels forest B's tips (if $\gamma > 0$). This is the primary mechanism of inter-forest interaction.
- Spatial exclusion: Inert sites from all forests occupy lattice positions. They cannot be grown into by any forest.
- No direct energy transfer: Forests do not share energy. Each forest's energy budget is independent (plus external input I).

B.6 Population-Level Energy Dynamics

Total energy across all forests:

$$E_{\text{pop}}(t) = \sum_{f \in \text{calF}_t} \sum_{p \in f} E(p, t)$$

Each REPRODUCE event changes total population energy by:

$$\Delta E_{\text{pop}} = -\epsilon_{\text{reproduce}} + (\text{establishment costs of new forest})$$

The overhead $\epsilon_{\text{reproduce}}$ is a pure loss. The new seed's energy E_{seed} is transferred from parent to offspring (zero net change). But the offspring must spend energy on EXTEND and SENSE before it has enough tips to collect external input efficiently. This establishment phase is energy-negative, creating a structural deficit that no parameter setting closes.

B.7 Empirical Characterization

Experimental design: 3 initial seeds ($P_{\text{bifurcate}} = 8, 14, 22$), shared trail map, reproduction enabled with moderate threshold, mutation $\sigma = 1.0$ on $P_{\text{bifurcate}}$. Systematic variation of external energy input I , reproduction threshold, mutation rate, trail avoidance, and inert site decay. Additional evolutionary replicates with selection over extended timescales.

Key findings:

Test Result

Reproduction mechanism Works as specified. Viable independent seeds created.

Heritable variation Confirmed. $P_{\text{bifurcate}}$ distribution diversifies over generations.

Selection Confirmed. Populations drift toward parameters that improve short-term fitness (moderate $P_{\text{bifurcate}}$, easier branching).

Population persistence at $I \leq 5.0$ Never observed. All runs freeze.

Population persistence at $I = 10, 20, 50$ Never observed. Birth/death ratio scales with I but asymptotes below 1.0.

Maximum-permissive configuration ($I=50, \gamma=0$, inert decay, no mutation, very low reproduction threshold) Freezes at $t=874$. Birth/death ratio ~ 0.97 . Peak simultaneous forests: 19.

Evolutionary replicates (selection + variation, $I=50$, 5 replicates) All freeze. Best birth/death ratio: 0.98. Selection improves fitness but cannot cross the structural bound.

Interpretation: The reproduction transaction has an irreducible energy cost. Parent sacrifices productive position + pays overhead. Offspring starts from single seed with establishment costs. The birth/death ratio approaches 1.0 asymptotically as external input increases, but the asymptote is below 1.0. This is a structural property of the transaction design, not a parameter-tuning problem.

B.8 Relationship to the Eventuality Theorem

The individual-forest eventuality theorem (Section 13) guarantees each forest freezes in finite time without external energy. With reproduction, the population-level energy $E_{\text{pop}}(t)$ decreases at each REPRODUCE event by at least $\epsilon_{\text{reproduce}}$, plus the establishment deficit of the offspring forest. The total number of REPRODUCE events across all forests is bounded by $E_{\text{pop}}(0) / \epsilon_{\text{reproduce}}$. After the last reproduction event, each remaining forest freezes individually.

External energy input $I(p,t) > 0$ slows the decline but does not reverse it. Empirically, even at $I = 50$ per tip per step (net positive energy for SENSEing tips), the population eventually freezes because the reproduction overhead and establishment deficit exceed the marginal benefit of additional tips.

B.9 Conditions for Persistence (Conjectured)

The structural bound could potentially be overcome by mechanisms that bypass the establishment deficit:

1. Lateral spread (runners/stolons): Parent remains alive and connected to offspring. No establishment phase—the new site starts with access to the parent's energy network.
2. Multi-tip seeds: Offspring inherit a small initial structure (3–5 connected sites) rather than a single seed, bypassing the single-tip establishment bottleneck.
3. Inter-forest energy transfer: Parent forest subsidizes offspring for N steps after reproduction, covering the establishment cost.

These are beyond the current framework scope. They represent natural directions for future extensions, each with testable predictions about the birth/death ratio crossing 1.0.

B.10 Usage Guidance

Enable reproduction when modeling:

- Seasonal or opportunistic dispersal (dandelion seeds, fungal spores)
- Life-history strategies where individuals reproduce then die
- Populations where heritable variation and selection are of interest
- Systems where the extended active phase and multi-forest diversity add explanatory value

Do not expect reproduction to produce:

- Indefinitely persistent populations
- Ecosystem steady states
- Birth/death ratios ≥ 1.0

Reproduction enriches the framework's behavioral repertoire within its fundamental thermodynamic constraints. It does not break those constraints.

Appendix C: Parameter Boundaries and Degenerate Limits

C.1 Hard Invariants

The following strict inequalities must hold for the framework to produce meaningful behavior and for the Eventuality Theorem to apply:

Invariant Consequence of Violation

$\epsilon_{\text{extend}} > 0$ Termination proof fails; infinite growth possible
 $\epsilon_{\text{sense}} > 0$ Infinite SENSE loops; never RESTRICT or EXTEND
 $\beta < 1$ (strict) CONSERVE loses no energy; potential oscillations
 $\iota \geq 1$ Co-location breaks conflict resolution and gradient regularization
 $R \geq 5$ No meaningful neighborhood; gradient always zero; no EXTEND

C.2 Degenerate Limit Behavior

Systematic testing at parameter extremes reveals predictable failure modes:

Parameter Degenerate Value Outcome Failure Mode

$\backslash\text{varepsilon}_{\text{extend}}$ 0 Infinite growth (or safety cap) Termination proof fails
 $\backslash\text{varepsilon}_{\text{sense}}$ 0 Infinite SENSE loops Non-termination
 $\backslash\beta$ 1.0 Perfect recycling, delayed freeze Oscillatory potential
 $\backslash\gamma \geq 10$ Immediate repulsion, tiny structures Over-constrained; no exploration
 $\backslash\iota$ 0 Division by zero risk, immediate RESTRICT everywhere Computational failure
 R 0 Zero gradient, immediate stall No sensing possible
 $\backslash\tau$ 0 Massive uncontrolled growth No energy-based death
 $P_{\text{bifurcate}}$ 0 Hyper-bushy; loses all depth Meaningless for most domains
 $P_{\text{bifurcate}} \geq 1000$ Strictly linear; never branches Loses regime diversity

C.3 Meaningful Parameter Ranges

The framework produces biologically or structurally plausible behavior within these ranges. Outside these ranges, behavior becomes either trivial (immediate freeze) or degenerate (non-terminating or computationally unstable):

Parameter Meaningful Range

$\backslash\text{varepsilon}_{\text{extend}}$ 3–30
 $\backslash\text{varepsilon}_{\text{sense}}$ 1–8
 $\backslash\beta$ 0.6–0.95
 $\backslash\gamma$ 0–4
 $P_{\text{bifurcate}}$ 5–100
 $\backslash\tau$ 3–25
 $\backslash\iota$ 1–5
 R 10–50

C.4 Robustness Assessment

The framework remains well-behaved across approximately 2–3 orders of magnitude in most parameters. Even at extremes, the synchronous update and pressure/energy mechanics prevent crashes in most cases. The design is defensively solid—failures are predictable, bounded, and match the theoretical analysis.

Appendix D: Computational Scaling

D.1 Theoretical Complexity

All operations per timestep are bounded by the finite support invariant (I5):

- Neighborhood computation: $O(N_{\text{active}} \cdot N_{\max}(R))$ naive, $O(N_{\text{active}})$ with spatial indexing
- Decision computation: $O(N_{\text{active}})$ — each site evaluates its own decision independently
- Conflict resolution: $O(C \log C)$ where C is the number of candidate positions
- Trail update: $O(S)$ where S is the support size of the trail map

Phase 1 (decisions) and Phase 4b (conflict resolution within spatial buckets) are embarrassingly parallel.

D.2 Practical Scaling (Naive Implementation)

Measured on a Python prototype without spatial indexing:

Active Sites Time per Step

100 ~0.02s

1,000 ~0.65s
5,000 ~15.6s
10,000+ >1 minute

D.3 Practical Scaling (Optimized Implementation)

With spatial hashing (3D grid buckets of size $\approx R$ or 1):

Active Sites Time per Step

1,000 <0.01s
10,000 ~0.05s
50,000 ~0.5s
100,000+ Feasible on GPU/distributed

D.4 Required Data Structures

For efficient implementation:

- Site positions: 3D spatial hash map (bucket size = R) for $O(1)$ neighborhood queries
- Trail map: Sparse dictionary (hash map); only visited positions stored
- Candidate positions: Temporary spatial map for Phase 4b conflict detection
- CONSERVE buffer: Per-site accumulator for Phase 2 $\rightarrow$ Phase 5 transfer

D.5 Memory Scaling

- Active sites: $O(N_{\text{active}})$
- Trail map: $O(\text{distinct positions visited})$ — grows with explored volume, stabilizes under decay
- Inert sites: $O(\text{total sites ever created})$ — can be pruned if inert site decay is enabled

D.6 Long-Timescale Stability

- All accumulators (energy, pressure, trail) are rational numbers
- No overflow with 64-bit integers or double-precision floats for realistic runs ($<10^6$ timesteps)
- Trail saturation under decay: equilibrium $\approx \Delta_{\text{trail}} / \Delta_{\text{decay}}$ (e.g., $0.5/0.01 = 50$)
- Pressure bounded by reset on RESTRICT/CONSERVE and cap at P_{erode}

D.7 Scaling Limits Summary

Scale Active Sites Feasible? Bottleneck

Single neuron <5,000 Excellent (naive) None

Neural population 10k–100k Good (spatial index) Memory + conflict resolution

Cortical column 100k–1M Feasible (optimized) Memory bandwidth

Whole brain slice 1M+ Heavy (distributed/GPU) Inter-node communication

The specification imposes no theoretical barriers to arbitrary scale. Practical limits are standard spatial simulation engineering challenges.

Scope and Status

Natural Math is a formal modeling language for acyclic spatial growth processes under finite constraints. Its strongest theoretical result is the eventual inactivity theorem for closed systems, which extends empirically to populations of reproducing forests. Its value lies in providing a common, testable structure—the four decision states, local agency, energy accounting, lifecycle, regime taxonomy, and reproduction extension—applicable across domains that share the four defining properties.

What has been empirically validated (~40 runs, six test suites):

- Core microscopic rules produce consistent, predictable behavior
- Eventuality theorem holds for all closed-system configurations and has been verified through direct energy accounting
- Three primary behavioral regimes emerge from parameter variation and are predictable from inputs alone (91% accuracy) and recoverable from morphology (~90–95% accuracy)
- Reproduction works as a mechanism and produces heritable variation under selection, but does not produce indefinite population persistence
- The birth/death ratio asymptotes below 1.0—a structural property of the reproduction transaction
- Framework behavior is robust to adversarial environments (checkerboard, funnel, maze, all-obstacle, periodic boundaries)
- Degenerate parameters produce exactly the failures predicted by the hard invariants
- Energy is strictly conserved across all transactions
- The framework scales mathematically to arbitrary size and practically to biologically relevant scales with standard optimizations
- Natural Math produces qualitatively and quantitatively superior structures compared to null models (random walks)

What remains open:

- Per-regime depth models (mechanisms identified, equations partially fitted)
- Branch reactivation (lateral tip spawning from established branches)
- Persistence mechanisms (lateral spread, multi-tip seeds, inter-forest energy transfer)
- Domain-specific gradient fields (gravity, water potential, chemical gradients)
- Domain-specific target-triggered state transitions

Domains validated:

- Root growth (botany)
- Urban development (infrastructure)
- Fungal mycelium (mycology)
- Neural arborization (neuroscience)

Version 2.3 is internally consistent, empirically characterized, and implementable. The framework is ready for domain-specific application, open-source implementation, and the next generation of extensions that address the structural limits documented here.

End of Version 2.3 — Final